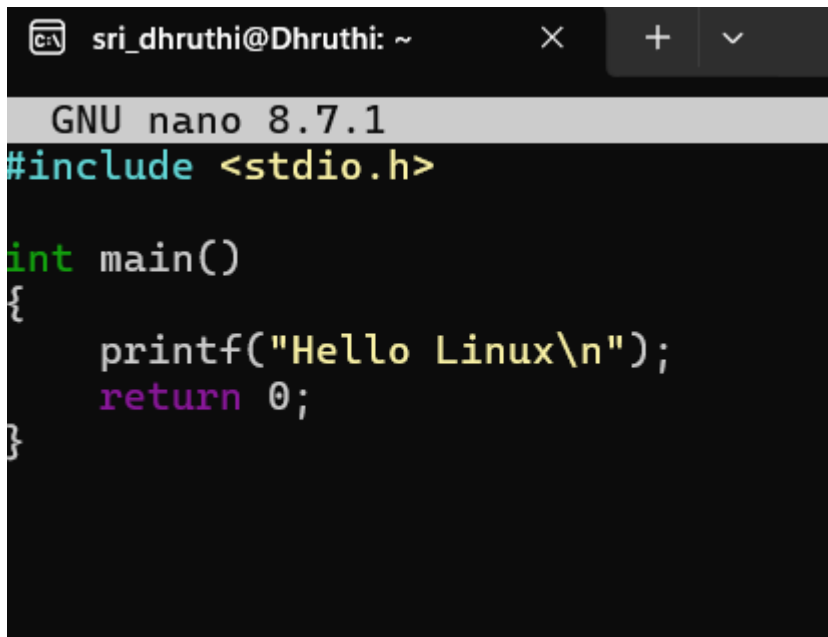


OS UBUNTU PROGRAMS

WEEK -1

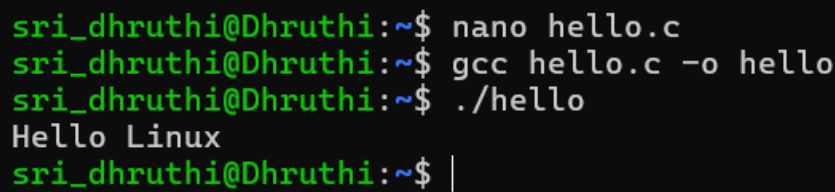
PROGRAM-1



```
sri_dhruthi@Dhruthi: ~
GNU nano 8.7.1
#include <stdio.h>

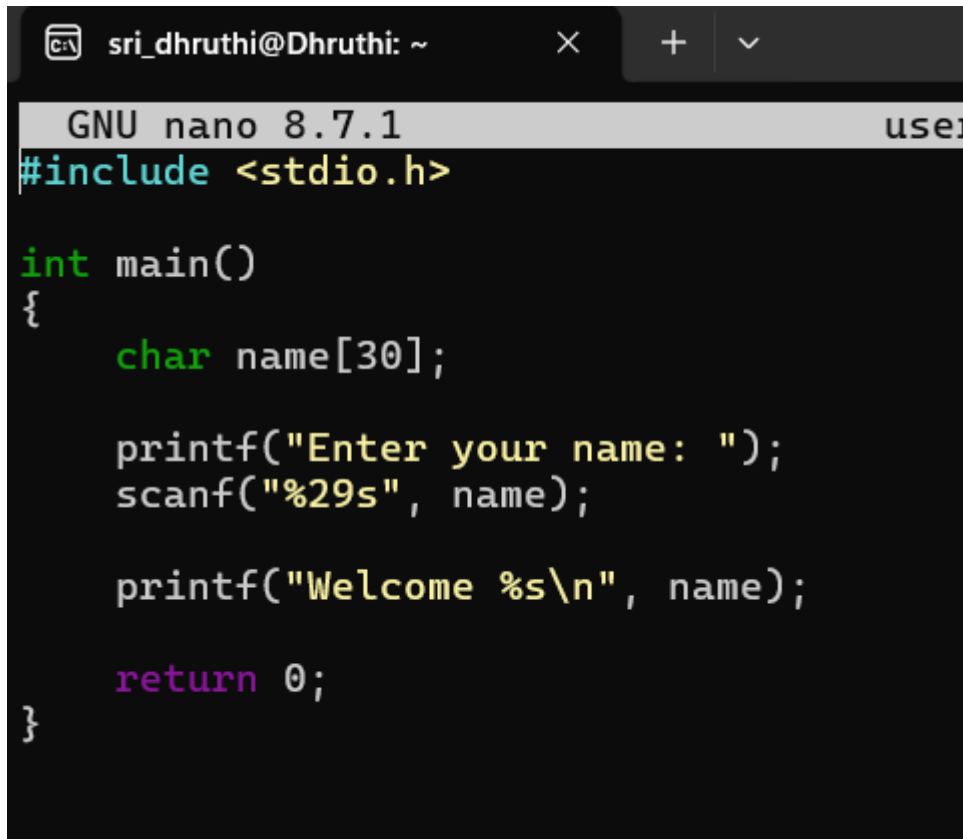
int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

OUTPUT:



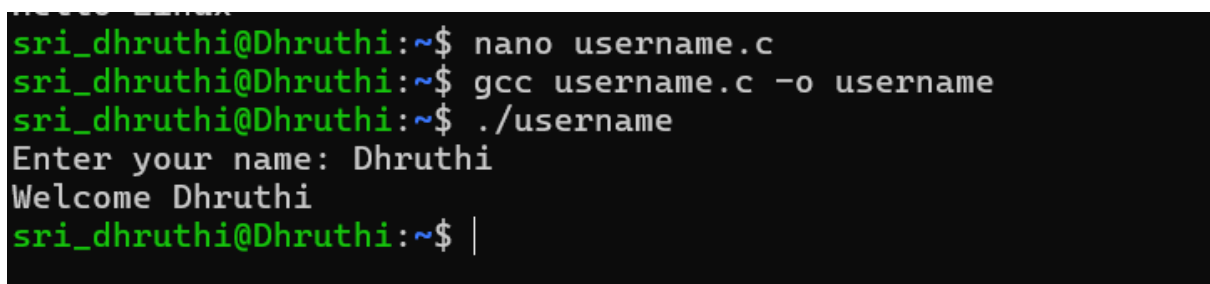
```
sri_dhruthi@Dhruthi:~$ nano hello.c
sri_dhruthi@Dhruthi:~$ gcc hello.c -o hello
sri_dhruthi@Dhruthi:~$ ./hello
Hello Linux
sri_dhruthi@Dhruthi:~$ |
```

PROGRAM 2



```
sri_dhruthi@Dhruthi: ~  
GNU nano 8.7.1 username.c  
#include <stdio.h>  
  
int main()  
{  
    char name[30];  
  
    printf("Enter your name: ");  
    scanf("%29s", name);  
  
    printf("Welcome %s\n", name);  
  
    return 0;  
}
```

OUTPUT:



```
sri_dhruthi@Dhruthi:~$ nano username.c  
sri_dhruthi@Dhruthi:~$ gcc username.c -o username  
sri_dhruthi@Dhruthi:~$ ./username  
Enter your name: Dhruthi  
Welcome Dhruthi  
sri_dhruthi@Dhruthi:~$ |
```

PROGRAM 3

```
GNU nano 8.7.1 pid.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process ID = %d\n", getpid());

    return 0;
}
```

```
Welcome Dhruthi
sri_dhruthi@Dhruthi:~$ nano pid.c
sri_dhruthi@Dhruthi:~$ gcc pid.c -o pid
sri_dhruthi@Dhruthi:~$ ./pid
Process ID = 627
```

PROGRAM 4

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();

    printf("Hello\n");

    return 0;
}
```

OUTPUT:

```
sri_dhruthi@Dhruthi:~$ nano fork.c
sri_dhruthi@Dhruthi:~$ gcc fork.c -o fork
sri_dhruthi@Dhruthi:~$ ./fork
Hello
Hello
```

PROGRAM 5

```
GNU nano 8.7.1 parent_chi
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("I am Child Process\n");
    }
    else
    {
        printf("I am Parent Process\n");
    }

    return 0;
}
```

```
sri_dhruthi@Dhruthi:~$ nano parent_child.c
sri_dhruthi@Dhruthi:~$ gcc parent_child.c -o parent_child
sri_dhruthi@Dhruthi:~$ ./parent_child
I am Parent Process
I am Child Process
sri_dhruthi@Dhruthi:~$
```

PROGRAM 6

```
GNU nano 8.7.1 display_pid.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else
    {
        printf("Parent PID = %d\n", getpid());
    }

    return 0;
}
```

```
sri_dhruthi@Dhruthi:~$ nano display_pid.c
sri_dhruthi@Dhruthi:~$ gcc display_pid.c -o display_pid
sri_dhruthi@Dhruthi:~$ ./display_pid
Parent PID = 738
Child PID = 739
Parent PID = 318
```

PROGRAM 7

```
sri_dhruthi@Dhruthi: ~  GNU nano 8.7.1  wait_example.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("Child Running\n");
    }
    else
    {
        wait(NULL);

        printf("Parent Running After Child\n");
    }

    return 0;
}
```

```
sri_dhruthi@Dhruthi:~$ nano wait_example.c
sri_dhruthi@Dhruthi:~$ gcc wait_example.c -o wait_example
sri_dhruthi@Dhruthi:~$ ./wait_example
Child Running
Parent Running After Child
sri_dhruthi@Dhruthi:~$ nano exec_example.c
```

PROGRAM 8

```
GNU nano 8.7.1                                exec
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Before exec()\n");

    execlp("ls", "ls", NULL);

    printf("After exec()\n");

    return 0;
}
```

```
sri_dhruthi@Dhruthi:~$ nano exec_example.c
sri_dhruthi@Dhruthi:~$ gcc exec_example.c -o exec_example
sri_dhruthi@Dhruthi:~$ ./exec_example
Before exec()
display_pid      filename      hello        pid          wait_example
display_pid.c    filename.c    hello.c      pid.c        wait_example.c
exec_example     fork         parent_child username
exec_example.c   fork.c       parent_child.c username.c
```

PROGRAM 9

```
GNU nano 8.7.1                                fork_exec.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        execlp("date", "date", NULL);
    }
    else
    {
        wait(NULL);

        printf("Child Completed\n");
    }

    return 0;
}
```

```
sri_dhruthi@Dhruthi:~$ nano fork_exec.c
sri_dhruthi@Dhruthi:~$ gcc fork_exec.c -o fork_exec
sri_dhruthi@Dhruthi:~$ ./fork_exec
Tue Aug  4 07:13:49 UTC 2026
Child Completed
```